# Neeraj Sahu

R&D Professional — Analytics in Wind Energy
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## Skills

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| **Technical Skills:** | Performance Validation, Wind Farm Control, Certification |
| **Languages:** | Python, MATLAB, Bash, R |
| **Simulation & Modelling:** | BHaWC, FLORIS, PyMC (Bayesian Modeling), State-Space Models |
| **Standards:** | IEC 61400 |
| **Tools:** | Git, LaTeX, Jira |

## Experience

**Siemens Energy Wind Power** Solution Designer - R&D *July 2021 – Present*

- **Quarterly Controller Certification:** Owned the planning and execution of quarterly certification documentation for the D8, SG5X, and NG-X Controllers, supervising with a senior consultant and intern to meet deadlines.
  - Aligned with multiple stakeholders such as Solution Owners, Product Owners, and Tool Developers.
  - Evangelized a comprehensive understanding of the certification process, including the IEC 61400 standard and its application to wind turbine control systems.
  - Developed a robust framework for the certification process, ensuring that all necessary documentation and testing procedures were in place.
  - Implemented a rigorous testing and validation process to ensure that all control systems met the required standards and specifications.
- **Wind Farm Control:** Developed and implemented advanced control algorithms for wind farm optimization, resulting in improved energy production and reduced maintenance costs.
  - Conducted extensive simulations and modeling to evaluate the performance of control strategies under various operating conditions.
  - Collaborated with cross-functional teams to integrate control systems with existing infrastructure and ensure seamless operation.
- **Performance Validation:** Developed a method to quantify the power benefit of wake steering in offshore wind farms using only existing SCADA data, without the need for external measurement tools.
  - Implemented uncertainty quantification and optimization techniques to improve the accuracy and reliability of the wake steering power benefit measurements.
- **Certification:** Developed a comprehensive understanding of the IEC 61400 standard and its application to wind turbine control systems.
  - Implemented a rigorous testing and validation process to ensure that all control systems met the required standards and specifications.
- **Collaboration and Communication:** Collaborated with cross-functional teams, including Solution Owners, Product Owners, and Tool Developers, to ensure that all aspects of the certification process were aligned and integrated.
  - Communicated complex technical concepts and findings to both technical and non-technical stakeholders, ensuring that all parties had a clear understanding of the certification process and its outcomes.

## Industry Mention @ IEA Wind Task 44

**An industry-oriented method to measure Wake Steering Power Benefit in Offshore wind farms**

*Click here for YouTube presentation.*

- Long-term validation using only existing SCADA data without external measurement tools.
- Uncertainty quanitfication and optimization of wake steering power benefit.

## Education & Trainings

**Indian Institute of Technology, Madras**

*M.Tech. in Aerospace Engineering* *2019 – 2021*

- Statistical Estimation of Motion Parameters from Radar Signals of Manned Maneuvering Targets.
- Estimated motion parameters from radar-signal ensembles for a manned maneuvering aircraft.
- Implemented Kalman, Unscented Kalman, and Extended Kalman filters using time-series analysis, statistical estimation, optimization, and dynamical-system models.

**MITS (Govt), Gwalior, MP**
*BEng in Automobile Engineering* *2015 – 2019*

**Trainings and Courses**

*Technical Project Management* *Mar 2026*
*Critical Conversations and Conflict Management* *Aug 2025*